NATHANIEL POWELL

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github.com/Tophacks | personal-website-ten-lyart.vercel.app

EDUCATION

University of Waterloo Waterloo, ON
Bachelor of Science, Honours Mathematical Physics, Co-op Sep. 2025 - Present

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML/CSS
Full-Stack & Data: React, Next.js, Node.js, Express, FastAPI, PostgreSQL, Supabase, REST APIs, WebSockets
AI & Simulation: PyTorch, Stable-Baselines3, PPO, OpenCV, HOG/SVM, SUMO, TraCI
Engineering Tools: Git, GitHub Actions, Docker, Vite, Vercel, Figma, testing, data modeling, CI/CD

EXPERIENCE

Quadmates.ca Waterloo, ON
Founder & Full-Stack Developer | 2026 - Present

- Built and launched a student networking platform serving approximately 50 early users, developing a React frontend, Node.js/Express APIs, and PostgreSQL/Supabase data models.
- Implemented authentication, study-partner matching, student profiles, real-time messaging, and group workflows; prioritized reliability and UX improvements using product feedback.
- Optimized direct and group messaging performance, reducing latency by 70%; adapted the web application for an iOS prototype using Capacitor and Vite.

Infosys Limited - InStep Internship Program Bengaluru, India
Incoming Computer Engineering Assistant - AI Agent Engineering | Jan. - Apr. 2027

- Selected for the Infosys InStep international internship program in the Office of the CTO, with a planned focus on enterprise AI agent development and intelligent workflow automation.

UW Orbital Waterloo, ON
Business & Finance Team Member | Jan. 2026 - Present

- Negotiated corporate manufacturing and resource support for a satellite engineering program, resolving sponsor production-schedule conflicts through technical planning and compromise.
- Supported interdisciplinary engineering operations, stakeholder communication, and expense tracking; explored Python automation for financial-data processing.

PROJECTS

TransitOps | *Python, SUMO, TraCI, FastAPI, Next.js, WebSockets*

- Developed a transit-operations simulator comparing baseline and adaptive dispatch under surges and disruptions; connected SUMO/TraCI signal and traffic telemetry to FastAPI WebSockets and a browser dashboard.

NeuroSim 2.0 | *Python, PyTorch, Stable-Baselines3, Reinforcement Learning*

- Built a simplified rocket-landing environment for closed-loop control using PPO; compared policies by landing success, touchdown error, crashes, and fuel usage.

Computer Vision Security Camera | *Python, OpenCV, Tkinter*

- Implemented HOG/SVM person detection with a multithreaded producer-consumer pipeline, frame sampling, and bounded buffering to manage processing load.

Staffflow | *Full-Stack Web Application, Role-Based Access Control*

- Developed staff-management functionality with role-specific authorization workflows for managers, staff, and administrators.